

Lecture 2: Weierstrass Theorem

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1 Why existence needs a theorem

For a function $f : X \rightarrow \mathbb{R}$, the number

$$\inf_{x \in X} f(x)$$

is the greatest lower bound of the set of function values. Its existence as an extended real number does not mean that some point in X achieves it. For example, $f(x) = x$ has infimum 0 on $X = (0, 1)$, but no point of $(0, 1)$ has value 0. The Weierstrass theorem in Lecture 2 gives conditions under which an infimum and a supremum are actually attained.

Recall that a set $X \subseteq \mathbb{R}^n$ is compact if every sequence in X has a subsequence converging to a point of X . In Euclidean space, the Heine–Borel theorem says that this is equivalent to X being closed and bounded.

2 The Weierstrass theorem

Theorem 1 (Weierstrass extreme value theorem). *Let $X \subseteq \mathbb{R}^n$ be nonempty and compact, and let $f : X \rightarrow \mathbb{R}$ be continuous. Then f attains both its minimum and its maximum on X . Equivalently, there exist $x_{\min}, x_{\max} \in X$ such that*

$$f(x_{\min}) = \min_{x \in X} f(x) = \inf_{x \in X} f(x)$$

and

$$f(x_{\max}) = \max_{x \in X} f(x) = \sup_{x \in X} f(x).$$

The minimizing and maximizing points need not be unique.

3 Detailed proof

Proof. We first prove that a minimum is attained. The argument has three parts: show that the infimum is finite, construct a sequence approaching it, and use compactness to obtain a convergent subsequence.

Step 1: f is bounded below on X . Suppose, to the contrary, that f is not bounded below. For each positive integer k , choose $y_k \in X$ such that

$$f(y_k) < -k.$$

Compactness of X gives a subsequence (y_{k_j}) and a point $\bar{y} \in X$ such that $y_{k_j} \rightarrow \bar{y}$. Continuity at $\bar{y}$ implies

$$f(y_{k_j}) \rightarrow f(\bar{y}),$$

and the limit on the right is a finite real number. But $f(y_{k_j}) < -k_j$ forces $f(y_{k_j}) \rightarrow -\infty$, which is a contradiction. Thus f is bounded below. Since X is nonempty, the number

$$\alpha := \inf_{x \in X} f(x)$$

is therefore a finite real number.

Step 2: choose a minimizing sequence. By the definition of infimum, for every positive integer k there is a point $x_k \in X$ satisfying

$$\alpha \leq f(x_k) < \alpha + \frac{1}{k}. \quad (1)$$

Indeed, if no such point existed for a particular k , then $\alpha + 1/k$ would be a lower bound for $f(X)$ strictly larger than the greatest lower bound α . The squeeze theorem applied to Eq. (1) shows that

$$f(x_k) \longrightarrow \alpha.$$

The sequence (x_k) is called a minimizing sequence.

Step 3: compactness turns approximation into attainment. Since every x_k lies in the compact set X , there are indices $k_1 < k_2 < \dots$ and a point $x_{\min} \in X$ such that

$$x_{k_j} \longrightarrow x_{\min}.$$

Continuity of f gives

$$f(x_{k_j}) \longrightarrow f(x_{\min}).$$

The subsequence $(f(x_{k_j}))$ also converges to α , because the full sequence $(f(x_k))$ does. A real sequence has only one limit, so

$$f(x_{\min}) = \alpha = \inf_{x \in X} f(x).$$

Thus the infimum is attained and is a minimum.

Step 4: obtain the maximum. The function $-f$ is continuous on the same nonempty compact set X . Applying the minimum result to $-f$ gives some $x_{\max} \in X$ such that

$$-f(x_{\max}) = \min_{x \in X} (-f(x)).$$

Multiplying by -1 reverses the order and yields

$$f(x_{\max}) = \max_{x \in X} f(x).$$

This proves both assertions. □

4 Why each hypothesis matters

The assumptions in Theorem 1 cannot simply be dropped.

- **The set must be nonempty.** An empty feasible set contains no candidate that could attain a minimum or maximum.
- **Closedness matters.** On $X = (0, 1)$, the continuous function $f(x) = x$ has infimum 0, but $0 \notin X$.
- **Boundedness matters.** On $X = \mathbb{R}$, the continuous function $f(x) = x^2 - x^4$ is unbounded below. Also, $f(x) = e^{-|x|}$ has infimum 0, but $e^{-|x|} > 0$ for every $x \in \mathbb{R}$. These are the two nonexistence examples from Lecture 2.
- **Continuity matters.** On the compact set $X = [0, 1]$, define

$$f(0) = 1, \quad f(x) = x \quad \text{for } x \in (0, 1].$$

Then $\inf_{x \in X} f(x) = 0$, but the value 0 is never attained. The failure occurs because f is discontinuous at 0.

5 Consequences for unconstrained optimization

Lecture 2 also gives two common ways to use Weierstrass when the original domain is all of $\mathbb{R}^n$ and is therefore not compact.

Corollary 1 (Compact sublevel set). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous. Suppose there is a point $x_0 \in \mathbb{R}^n$ such that the sublevel set*

$$S_0 := \{x \in \mathbb{R}^n : f(x) \leq f(x_0)\}$$

is compact. Then f attains a global minimum on $\mathbb{R}^n$.

Proof. The set S_0 is nonempty because $x_0 \in S_0$. By Theorem 1, there is $x^* \in S_0$ such that

$$f(x^*) = \min_{x \in S_0} f(x).$$

If $x \notin S_0$, then $f(x) > f(x_0) \geq f(x^*)$. If $x \in S_0$, the definition of x^* gives $f(x) \geq f(x^*)$. Thus $f(x) \geq f(x^*)$ for every $x \in \mathbb{R}^n$, so x^* is a global minimizer. $\square$

Definition 1 (Coercive function). *A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is coercive if*

$$f(x) \longrightarrow +\infty \quad \text{as } \|x\|_2 \longrightarrow \infty.$$

Equivalently, for every $M \in \mathbb{R}$ there is an $R > 0$ such that $\|x\|_2 \geq R$ implies $f(x) > M$.

Corollary 2 (Coercivity). *Every continuous coercive function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ attains a global minimum.*

Proof. Choose any $x_0 \in \mathbb{R}^n$. By coercivity, there is $R > 0$ such that

$$\|x\|_2 \geq R \implies f(x) > f(x_0).$$

Consequently, the sublevel set $S_0 = \{x : f(x) \leq f(x_0)\}$ is contained in the ball of radius R and is bounded. It is also closed, because it is the inverse image of the closed set $(-\infty, f(x_0)]$ under the continuous map f . The Heine–Borel theorem therefore implies that S_0 is compact. Applying Corollary 1 proves that f has a global minimizer. $\square$

Coercivity guarantees existence, not uniqueness. Additional structure, such as strict convexity, is needed to conclude that the minimizer is unique.